The Jäschke–Skogestad temperature conjecture for heat-exchanger networks is false: disproof and corrected theorem

chemistry-open-problems · 18 October 2026 · independently verified

Solver: Claude Opus 5.5 (effort xhigh), sub-agent of the orchestrator; citing-works search by a nested read-only Claude Opus 5.5 sub-agent (effort xhigh) · LaTeX edition: Claude Opus 5.5 (effort xhigh) · Independent verification and corrections: Claude Opus 5.5 (effort xhigh), fresh context

Abstract. Jäschke and Skogestad (Comput. Chem. Eng. 2014) showed that optimal operation of a heat-exchanger network with stream splits only requires temperature measurements: with an arithmetic-mean temperature-difference (AMTD) model, the marginal cost of a split branch is claimed to be a sum $\sum_i p_{ij} a_{ij}$ of recursively defined "Jäschke temperatures". They proved this for branches with $M_j \le 2$ exchangers, reported a proof of $M_j = 3$ for constant hot-stream temperatures, and conjectured the general case. We show that the published recursion is wrong for every $M_j \ge 3$. Hence the conjecture is false, and so is the $M_j = 3$ case reported as proved. The error in that proof is located: the raw computer-algebra output printed in the paper is correct, but a manual simplification drops a term. An exact rational counterexample has the paper's controlled variable equal to zero while the true gradient is $-4972391424/1518467647$. We prove the corrected recursion $a_i = c_i(\theta_i + \theta_{i-1} - \sum_{l<i} a_l)$ for every M_j, with a half-page proof based on the affine structure of each exchanger. On the paper's own validation case, the published controlled variable vanishes at a split that differs from the AMTD optimum by 0.0026, so the published recursion alone produces a split error of the order the paper attributed to flatness and round-off.

1 Source and exact statement

J. Jäschke, S. Skogestad, *Optimal operation of heat exchanger networks with stream split: Only temperature measurements are required*, Computers and Chemical Engineering 70 (2014) 35–49, doi:10.1016/j.compchemeng.2014.03.020 [1] (Manfred Morari special issue). Read in full: the publisher PDF self-archived at https://skoge.folk.ntnu.no/publications/2014/jaschke-hen-paper-cce-morari-issue/impress.pdf, plus the older preprint .../old-version/HenCCE.pdf. Page numbers below are journal page numbers.

Model (Section 3.1, p. 37–38). N parallel branches split from a cold feed (temperature T_0, heat-capacity flow rate w_0); branch j carries $w_j = F_j c_{p0}$, $\sum_j w_j = w_0$, and M_j heat exchangers in series. For exchanger i on branch j: cold outlet $T_{i,j}$ (cold inlet $T_{i-1,j}$, $T_{0,j} = T_0$), hot inlet $T_{hi,j}$, hot outlet $T^{out}_{hi,j}$, hot heat-capacity flow rate $w_{hi,j}$, conductance $UA_{i,j}$:

$$Q_{i,j} = w_j(T_{i,j} - T_{i-1,j}) \quad (9), \qquad Q_{i,j} = w_{hi,j}(T_{hi,j} - T^{out}_{hi,j}) \quad (10),$$

$$Q_{i,j} = UA_{i,j}\,\frac{(T_{hi,j} - T_{i,j}) + (T^{out}_{hi,j} - T_{i-1,j})}{2} \quad (12,\ \text{Assumption 3, AMTD}).$$

(The printed (9)–(10) contain the index misprints w_i for w_j and $T^{out}_{i-1,j}$ for $T^{out}_{hi,j}$; the intended model is the one above, as Fig. 3 and (A.1)–(A.4) show.) Assumptions 1–2: single phase, constant heat capacities. Cost (14): $J = -\sum_j \sum_i p_{i,j} Q_{i,j}$. Shifted temperatures (15): $\theta = T - T_0$.

Theorem 2 (p. 39), verbatim. "*Under Assumptions 1–3, and equal prices $p_{i,j} = 1$ in the cost function (14), the marginal costs for each branch $j = 1 \dots N$ can be expressed as*

$$\frac{\partial J}{\partial w_j} = T_{J,j} \quad (16)$$

where $T_{J,j}$ is the Jäschke Temperature on branch j, defined as $T_{J,j} = \sum_{i=1}^{M_j} a_{i,j}$ (17), with the parameter $a_{i,j}$ defined recursively as

$$a_{i,j} = \frac{(\theta_{i,j} - \theta_{i-1,j})(\theta_{i,j} + \theta_{i-1,j} - a_{i-1,j})}{\theta_{hi,j} - \theta_{i-1,j}}, \qquad a_{0,j} = 0. \quad (18)"$$

Theorem 3 (p. 39) is the same with arbitrary prices: $\partial J / \partial w_j = T^e_{J,j} = \sum_i p_{i,j} a_{i,j}$ (19)–(20), with the same $a_{i,j}$.

The open statements, verbatim.

- p. 39, after Theorem 2: "*In Appendix A we provide a proof of Theorem 2 for $M_j = 1$ and $M_j = 2$. Moreover, we prove the case $M_j = 3$ for heat exchangers with constant hot stream temperatures (this corresponds to very large hot stream heat capacities). For $M_j \geq 4$ we conjecture that the Theorem is true.*" The same sentence is repeated after Theorem 3.

- p. 47, end of A.3: "*Although this expression was derived under the assumption of a constant hot stream temperature $T_{hi,j} = T^{out}_{hi,j}$, we conjecture that it is true for the case with a finite heat capacity (non-constant hot streams).*"

- p. 47, A.4 "Conjecture for the general case ($M \geq 4$)": "*Unfortunately it was computationally not feasible to perform the elimination for the case with $M_j > 3$. From the recurring pattern, that coefficients of the previous prices appear in a fixed structure in the coefficients of the next price, we conjecture the general formula for the case with more than three heat exchangers on a line as*" (A.14)–(A.16), identical to (18)–(20).

- p. 45, Conclusions: "*For the case with 1–3 heat exchangers per line, we have proven [...] that the Jäschke Temperatures are equal to the marginal costs. The more general case with more than 3 heat exchangers is conjectured.*"

- p. 49, Appendix B: "*In conclusion, the numerical results for this case study strongly suggest that the conjecture is true.*"

The recursion is unambiguous: (A.13) and (B.1) write it out for $M_j = 3, 4$, and each new term subtracts only the previous $a_{i-1,j}$; the efficiency form (29) of Section 6.1, $a_{i,j} = c_{i,j}(\theta_{i,j} + \theta_{i-1,j} - a_{i-1,j})$, says the same. The conjecture concerns Theorems 2 **and** 3, and covers $M_j \geq 4$ as well as $M_j = 3$ with finite hot heat capacities.

2 Results

Fix one branch and drop the index j. Write w for its cold heat-capacity flow rate, $\theta_0 = 0$, and $\Delta_i = \theta_i - \theta_{i-1}$. For each exchanger define

$$k_i = \frac{2}{UA_i} + \frac{1}{w_{h,i}} \quad (1/\infty := 0), \qquad c_i = \frac{\theta_i - \theta_{i-1}}{\theta_{h,i} - \theta_{i-1}}$$

(the paper's "efficiency", eq. 27).

Theorem A (corrected Jäschke temperature; all $M \geq 1$). Under Assumptions 1–3, for every exchanger i of the branch,

$$\frac{\partial Q_i}{\partial w} = \frac{a_i}{2}, \qquad a_i = c_i\Big(\theta_i + \theta_{i-1} - \sum_{l<i} a_l\Big),$$

where $\partial/\partial w$ holds T_0, the hot inlet temperatures, the $w_{h,i}$ and the UA_i fixed. Hence

$$\frac{\partial J_j}{\partial w_j} = -\tfrac{1}{2}\, T^{e*}_{J,j}, \qquad T^{e*}_{J,j} := \sum_{i=1}^{M_j} p_{i,j}\, a_{i,j}, \qquad \text{and for } p_{i,j} \equiv 1: \ \frac{\partial J_j}{\partial w_j} = -\tfrac{1}{2}\, T^{*}_{J,j},$$

where $T^{*}_{J,j} = A_{M_j}$ is given by the affine recursion

$$A_0 = 0, \qquad A_i = (1 - c_i)\, A_{i-1} + c_i\, (\theta_i + \theta_{i-1}).$$

This is the same form as the cold-stream recursion $\theta_i = (1 - c_i)\theta_{i-1} + c_i\theta_{h,i}$. The formula uses only the cold temperatures $\theta_0, \ldots, \theta_M$ and the hot inlet temperatures $\theta_{h,i}$. Because the factor $-\frac{1}{2}$ is common to all branches, equal marginal costs (Theorem 1 of the paper) is **exactly** equivalent to $T^{e*}_{J,1} = \cdots = T^{e*}_{J,N}$. In particular the qualitative claim of the paper's title (under AMTD, temperature measurements suffice to locate the optimal split) is true for every M_j; what fails for $M_j \geq 3$ is the specific recursion (18) conjectured by the authors.

Corollary B (the conjecture is false). Let a^P_i denote the paper's recursion (18), and $d_i = a^P_i - a_i$. Then $d_1 = d_2 = 0$ and $d_i = c_i\,(A_{i-2} - d_{i-1})$ for $i \geq 3$. In particular $d_3 = c_3\, a_1 = c_3\, \theta_1^2/\theta_{h,1}$, which is nonzero whenever exchangers 1 and 3 both transfer heat. Consequently:

(1) The paper's formula is correct for $M_j \leq 2$ up to the harmless common factor $-\frac{1}{2}$ (Section 4.6). So Examples 1–2 and Case studies 1–3 of the paper are unaffected.

(2) Theorem 3 (prices) is false for every $M_j \geq 3$, even up to any normalisation constant. Suppose $\sum_i p_i a^P_i = \lambda \sum_i p_i a_i$ for all prices. Then $a^P_i = \lambda a_i$ for every i. The case $i = 1$ ($a^P_1 = a_1 \neq 0$) forces $\lambda = 1$, but $a^P_3 = a_3 + c_3 a_1 \neq a_3$.

(3) Theorem 2 ($p \equiv 1$) is false for every $M_j \geq 3$. The mechanism is general. Take any branch 1 with $T^P_{J,1} := \sum_i a^P_i \neq T^*_{J,1}$ and $T^P_{J,1} > 0$, and a branch 2 with a single exchanger. For one exchanger $\theta_{1,2} = c\,\theta_{h1,2}$ with $c = 2/(1 + w_2 k_{1,2})$ independent of $\theta_{h1,2}$, so $T_{J,2} = c^2\theta_{h1,2}$; choosing $\theta_{h1,2} = T^P_{J,1}/c^2$ makes the paper's controlled variable $T^P_{J,1} - T_{J,2}$ exactly zero, while by Theorem A the true gradient is $\mathrm{d}J/\mathrm{d}w_1|_{w_2 = w_0 - w_1} = -\frac{1}{2}\,(T^*_{J,1} - T^P_{J,1}) \neq 0$. This is independent of any normalisation.

- $M_1 = 3$: Section 3.4 gives an exact rational, physically meaningful instance.

- $M_1 \geq 4$: append to branch 1 of that instance $M-3$ exchangers with conductance ε and hot inlet temperatures different from θ_3. As $\varepsilon \to 0^+$, $c_i = 2/(1 + w(2/\varepsilon + 1/w_{h,i})) \to 0$ for $i \geq 4$, so $d_i \to 0$ for $i \geq 4$ and $T^P_{J,1} - T^*_{J,1} = \sum_i d_i \to c_3 a_1 \neq 0$. For small $\varepsilon > 0$, re-tune $\theta_{h1,2}$ as above so that the paper's controlled variable is again exactly zero; the gradient is then nonzero.

- Direct instances: the verifier's script `verification/v2_exact.py` builds exact physical counterexamples with $M_1 = 4$ and $M_1 = 5$ (claims X3a, X3b), and the paper's own Appendix B data give an $M_1 = 4$ counterexample to Theorem 3 (Section 3.5, claim B3) and, with all prices set to 1, to Theorem 2.

The computer-algebra checks for $M = 3, \ldots, 8$ and the random tests for $M \leq 8$ find the difference $T^P_J - T^*_J$ to be a nonzero rational function, and nonzero in every draw.

(4) The case the paper reports as **proved**, $M_j = 3$ with constant hot temperatures, is also wrong as printed. The first line of the paper's (A.12), the raw Maple coefficient of $p_{3,1}$, equals our a_3 exactly. The "simplification" in the next line of (A.12) silently drops $-a_{1,1}$ (checked symbolically, claim S4, and independently by the verifier, claims P3–P6). The paper's elimination was right; the hand simplification, the resulting (A.13), and the extrapolated pattern (A.15) are wrong.

Proposition C (any exchanger law that is linear in temperatures; LMTD). Suppose every exchanger satisfies $\Delta_i = \varepsilon_i\,(\theta_{h,i} - \theta_{i-1})$, where the cold-side effectiveness ε_i depends on w and on the exchanger parameters but not on the temperatures. This holds for AMTD and for LMTD/ε–NTU models of any flow arrangement. Put $\varphi_i = w\,\partial\varepsilon_i/\partial w$. Then

$$\frac{\partial Q_i}{\partial w} = \Big(1 + \frac{\varphi_i}{\varepsilon_i}\Big)\Delta_i + \varepsilon_i\Big(\theta_{i-1} - \sum_{l<i} \frac{\partial Q_l}{\partial w}\Big).$$

- For AMTD, $\varphi = -\varepsilon(2-\varepsilon)/2$, so $1 + \varphi/\varepsilon = \varepsilon/2$; this gives Theorem A.

- For a counter-current exchanger with the logarithmic mean (LMTD), $\varepsilon = (1-E)/(1-RE)$ with $E = e^{-N(1-R)}$, $N = UA/w$, $R = w/w_h$, and $\varphi = -N\,\partial_N\varepsilon + R\,\partial_R\varepsilon$. Now ε and $R = (\theta_h - \theta_h^{out})/\Delta$ are fixed by the four terminal temperatures, and $N = \ln\big((1-R\varepsilon)/(1-\varepsilon)\big)/(1-R)$. So the exact LMTD marginal cost is again a function of temperatures only, but it needs the **hot outlet** temperatures as well.

- For condensing hot streams ($R = 0$), $\varphi = (1-\varepsilon)\ln(1-\varepsilon)$ and no hot outlet temperature is needed.

- Under AMTD the R-dependence cancels exactly: $\varepsilon = 2/(1 + 2/N + R)$ depends on w only through wk_i. This is why the AMTD formula needs no heat capacities and no hot outlet temperatures.

- Applied to an LMTD plant, the (corrected) AMTD formula is only an approximation. In our random tests its relative error on the branch marginal cost has median $\approx 9\%$, with large errors for very unbalanced exchangers, consistent with the paper's own discussion (numerical evidence only).

3 Proofs

3.1 Lemma 1 (each exchanger is an affine map of the cold temperature)

For given θ_{i-1}, the three equations (9), (10), (12) have the unique solution

$$\theta_i = \theta_{i-1} + c_i(\theta_{h,i} - \theta_{i-1}), \qquad c_i = \frac{2}{1+wk_i}, \qquad Q_i = w\,\Delta_i, \qquad \theta_{h,i}^{out} = \theta_{h,i} - Q_i/w_{h,i}.$$

For $w_{h,i} = \infty$, take $\theta_{h,i}^{out} = \theta_{h,i}$ and drop (10).

Proof. Substitute $\theta_{h,i}^{out} = \theta_{h,i} - Q_i/w_{h,i}$ and $\theta_i = \theta_{i-1} + Q_i/w$ into (12):

$$Q_i = UA_i\Big(\theta_{h,i} - \theta_{i-1} - \frac{Q_i}{2w} - \frac{Q_i}{2w_{h,i}}\Big) \iff Q_i\Big(\frac{1}{UA_i} + \frac{1}{2w} + \frac{1}{2w_{h,i}}\Big) = \theta_{h,i} - \theta_{i-1}.$$

The bracket is positive, so Q_i is unique. Dividing by w gives $\Delta_i = 2(\theta_{h,i} - \theta_{i-1})/(1+wk_i)$. □

Remarks.

- c_i is the temperature ratio c_i of Section 2 whenever $\theta_{h,i} \neq \theta_{i-1}$.
- $(UA_i, w_{h,i})$ enter only through k_i. So, on the cold side, an exchanger with finite $w_{h,i}$ is identical to a constant-hot-temperature exchanger with $UA' = (1/UA_i + 1/(2w_{h,i}))^{-1}$. This makes the paper's reduction to constant hot temperatures in A.3 rigorous, and shows that the conjectured extension "to finite heat capacity" adds nothing: whatever holds for constant hot temperatures holds for finite ones.
- The paper's (A.5), $\theta_j = \theta_{hj}/(1 - w_j(1/w_{hj} + 2/UA_j))$, has a sign error and a missing factor 2. Section 4.6 explains why this does not affect the controlled variable.

3.2 Lemma 2

$$w\,\frac{\partial c_i}{\partial w} = -\,\frac{c_i(2-c_i)}{2}\,.$$

Proof. $\partial c_i/\partial w = -2k_i/(1+wk_i)^2$, and $wk_i = (2-c_i)/c_i$, so $w\,\partial c_i/\partial w = -2\,\frac{2-c_i}{c_i}\,\frac{c_i^2}{4} = -\frac{c_i(2-c_i)}{2}$. □

3.3 Proof of Proposition C (and so of Theorem A)

Let $D_i = w\,\partial\theta_i/\partial w$, with $D_0 = 0$ because T_0 is fixed.

Step 1 (cumulative balance). Summing (9) over $l \le i$ gives $\sum_{l\le i} Q_l = w\theta_i$. Differentiating,

$$\sum_{l\le i} \frac{\partial Q_l}{\partial w} = \theta_i + D_i \qquad (i = 0, \dots, M). \tag{$*$}$$

Step 2 (chain rule through the affine maps). From $\theta_i = \theta_{i-1} + \varepsilon_i(\theta_{h,i} - \theta_{i-1})$ with $\theta_{h,i}$ fixed,

$$D_i = (1-\varepsilon_i)D_{i-1} + \varphi_i(\theta_{h,i} - \theta_{i-1}) = (1-\varepsilon_i)D_{i-1} + \frac{\varphi_i}{\varepsilon_i}\,\Delta_i.$$

For $\theta_{h,i} = \theta_{i-1}$, read $\frac{\varphi_i}{\varepsilon_i}\,\Delta_i$ as $\varphi_i(\theta_{h,i} - \theta_{i-1}) = 0$.

Step 3. Taking the difference of $(*)$ at i and $i-1$,

$$\frac{\partial Q_i}{\partial w} = \Delta_i + D_i - D_{i-1} = \Big(1 + \frac{\varphi_i}{\varepsilon_i}\Big)\Delta_i - \varepsilon_i D_{i-1}.$$

Eliminating D_{i-1} with $(*)$ at $i-1$, namely $D_{i-1} = \sum_{l<i} \partial Q_l/\partial w - \theta_{i-1}$, gives Proposition C.

AMTD. With $\varepsilon_i = c_i$, Lemma 2 gives $1 + \varphi_i/c_i = c_i/2$, so

$$\frac{\partial Q_i}{\partial w} = \frac{c_i}{2}\Big(\Delta_i + 2\theta_{i-1} - 2\sum_{l<i}\frac{\partial Q_l}{\partial w}\Big) = \frac{c_i}{2}\Big(\theta_i + \theta_{i-1} - \sum_{l<i} 2\,\frac{\partial Q_l}{\partial w}\Big).$$

Setting $a_i := 2\,\partial Q_i/\partial w$ gives exactly the recursion of Theorem A, which determines the a_i uniquely from a_1 on. The statements about J_j follow from $J_j = -\sum_i p_{i,j} Q_{i,j}$. Since $A_i = \sum_{l\le i} a_l$, we get $A_i = A_{i-1} + c_i(\theta_i + \theta_{i-1} - A_{i-1})$, which is the affine form. □

Corollary B, the recursion for d_i. $a_i^P - a_i = c_i(\sum_{l<i} a_l - a_{i-1}^P) = c_i(A_{i-2} + a_{i-1} - a_{i-1}^P) = c_i(A_{i-2} - d_{i-1})$, with $d_1 = 0$ (both equal $c_1\theta_1$) and $d_2 = c_2(A_0 - d_1) = 0$.

3.4 Exact counterexample ($M_1 = 3$, $M_2 = 1$, prices 1)

Produced by `scripts/counterexample_exact.py` (which also checks it and rewrites `certificates/counterexample.json`); the JSON file is re-checked from its raw parameters alone by the verifier's `verification/v2_exact.py` (claims X1a–X1f). All values are exact rationals.

Data.

- $T_0 = 20$ °C, $w_0 = 100$ kW/K, split $w_1 = w_2 = 50$ kW/K.
- Branch 1: $UA = (56, 34, 28)$ kW/K, $w_h = (88, 51, 78)$ kW/K, $\theta_h = (48, 52, 60)$ K, i.e. hot inlets at 68, 72 and 80 °C.
- Branch 2: $UA_{1,2} = 200$, $w_{h1,2} = 50$, and

  $$\theta_{h1,2} = \frac{13184889712830714450}{136131266865364681} \approx 96.854 \text{ K}.$$

  This value is chosen as $T_{J,1}^P/c^2$ with $c = 4/5$ (Corollary B, item 3).

What the operating point satisfies.

- It is physical: in every exchanger the cold stream is heated, the cold outlet stays below the hot inlet, the hot outlet stays above the cold inlet, and $0 < c_i < 1$. The largest ratio of terminal temperature differences is 1.64, so in every exchanger AMTD exceeds LMTD by at most about 2% (the model itself is AMTD by Assumption 3).
- Branch-1 temperatures: $\theta_1 \approx 28.623$, $\theta_2 \approx 38.123$, $\theta_3 \approx 46.517$.
- The paper's controlled variable vanishes exactly: $T_{J,1} - T_{J,2} = 0$, with $T_{J,1} = T_{J,2} \approx 61.98671$.
- The exact gradient of the recovered heat, computed by exact forward-mode differentiation of the raw model (no formula used), is

  $$\frac{\mathrm{d}Q_{tot}}{\mathrm{d}w_1}\Big|_{w_2 = w_0 - w_1} = -\frac{4972391424}{1518467647} \approx -3.2746 \neq 0.$$

  This equals $(T_{J,1}^* - T_{J,2}^*)/2 = -c_3 a_1/2$, with $T_{J,1}^* \approx 55.43749$.

- The split $w_1' = 99/2$ recovers strictly more heat: by about 1.47 kW, exactly.

So the point where the paper's controller settles is not optimal. Numerically (illustration only), the true optimum is $w_1^\star \approx 47.504$ ($u^\star \approx 0.4750$ instead of 0.5), and the loss is ≈ 4.08 kW (0.066% of 6204 kW). The loss is small because the optimum is flat; the split error is about 5% of w_0.

3.5 The paper's own validation case (Appendix B)

`scripts/reproduce_appendixB.py` (and, independently, `verification/v4_appendixB.py`) uses Table B.6: $M_1 = 4$, $M_2 = 1$, prices 1, 1.2, 1.3, 1.5 / 1.5. All functions are exact rational functions of $u = w_1/w_0$, and real roots are isolated exactly.

- (B1) The corrected controlled variable equals $-\frac{2}{w_0}\,\mathrm{d}J/\mathrm{d}u$ identically.
- (B2) $\mathrm{d}J/\mathrm{d}u$ has a unique root in $(0,1)$: $u_{opt} \in [0.7141194663, 0.7141194665]$, with $J = -3938.8167$ and $T_{end} = 158.77375$ °C. This matches the paper's "Optimized" column (0.7141, −3.9388, 158.7738).
- (B3) The paper's controlled variable (B.1), with the misprints of the printed (B.1) corrected, vanishes at $u_P \in [0.7167062190, 0.7167062200] \neq u_{opt}$, where $\mathrm{d}J/\mathrm{d}u \neq 0$. This is an exact $M_1 = 4$ counterexample to Theorem 3 on the paper's own data. Its only other root in $(0,1)$, $u \approx 0.0028$, lies in the unphysical AMTD region $c_{1,1} > 1$. With all prices set to 1 (Theorem 2), the paper's controlled variable vanishes at $u \approx 0.72505$ while the optimum is at $u \approx 0.72267$ (verifier's claim B4).

The paper reports $u = 0.7152$ for "Equal Jäschke Temp.". We could not reproduce this number from (B.1), with or without the printing typos ($\theta_{h1,1}^2$ for $\theta_{1,1}^2$, and $\theta_{h3,1} - \theta_{2,1}$ for $\theta_{h2,1} - \theta_{1,1}$ inside the $p_{3,1}$ term; with the typos the root moves to $u \approx 0.904$). The reported end temperature 158.7741 °C is what the AMTD model gives at $u = 0.7152$, but neither the paper's controlled variable (value ≈ 0.168) nor the corrected one (≈ -0.119) vanishes there, so the origin of the value 0.7152 is unknown. What the exact computation does show is that the paper's own controlled variable, evaluated without any round-off, vanishes 0.0026 away from the optimum. A split discrepancy of this order is therefore a consequence of the recursion error (Corollary B), not of round-off; the paper attributed the discrepancy in end temperature to "the flatness of the optimum and numerical round-off errors" and concluded that the results "strongly suggest that the conjecture is true". The cost function is too flat at the optimum (it changes by 5×10^{-6} relative between u_{opt} and u_P) to discriminate between the two formulas.

4 Verification summary, scope and limitations

4.1 Computer algebra (`scripts/verify_symbolic.py`, python-flint)

- The raw model of Section 1 is solved by Cramer's rule in $\mathbb{Q}(w, UA_i, w_{h,i}, \theta_{h,i})$ with fully generic symbols, and $\partial Q_i/\partial w$ is computed symbolically.
- **S1:** it equals $a_i/2$ (corrected recursion) as an identity of rational functions for every $i \leq M$. This is checked for $M = 1, \ldots, 5$ with finite $w_{h,i}$, all symbols generic (up to 16 indeterminates), and for $M = 1, \ldots, 8$ with $w_{h,i} = \infty$.
- **L1:** a single exchanger with a generic inlet temperature gives the same outlet as a constant-hot-temperature exchanger with $UA' = (1/UA + 1/(2w_h))^{-1}$. This is the

symbolic form of the remark after Lemma 1. Because UA' does not depend on w, the $w_{h,i} = \infty$ checks carry over to finite $w_{h,i}$ for all $M \leq 8$.

- The raw finite-w_h model at $M = 6$ (19 indeterminates) exceeded the 4 GB memory cap after passing S0 and S1 for $i \leq 5$ (`certificates/verify_symbolic_M6_raw_partial.log`).
- **S2, S3:** the paper's recursion is identical for $i \leq 2$, which is a symbolic identity. It differs for every $i \geq 3$, its branch total differs from $2\,\partial \sum_i Q_i/\partial w$, and $d_3 = c_3 a_1$. Non-vanishing is certified by exact evaluation at a rational point where no denominator vanishes.
- **S4:** the paper's printed Maple numerator in (A.12) equals the corrected a_3.
- These checks are redundant with the proof of Section 3, which does not rely on them.

4.2 Numerical evidence (`scripts/random_fd_tests.py`)

This is independent evidence, not the proof.

- 60-digit central finite differences on random data, $M = 1..8$, with random prices: the corrected formula agrees to relative error $< 10^{-30}$.
- The paper's T_J is off by at least 0.28% in every draw with $M \geq 3$.
- The LMTD form of Proposition C agrees to $< 10^{-25}$.

4.3 Independent verification

A verifier with a fresh context (no access to the solver's reasoning) re-derived Lemma 1, Lemma 2 and the recursion by hand, read the source paper (text and 400-dpi renderings of pp. 38–39 and 45–48), and wrote its own scripts in `verification/`, sharing no code with `scripts/`:

- `v1_symbolic.py` (sympy): each exchanger's equations solved by `sympy.solve`, derivative by `sympy.diff`; corrected recursion exact as an identity for $M \leq 3$ with finite and $M \leq 4$ with infinite hot capacities; published recursion wrong at $i = 3$; the factor $-\frac{1}{2}$ for $M = 1$.
- `v2_exact.py` (exact dual numbers, each exchanger solved as a 3×3 linear system): 320 random exact tests for $M \leq 8$; re-check of `certificates/counterexample.json` from its raw parameters; new exact physical counterexamples for $M_1 = 3$ with constant hot temperatures and for $M_1 = 4, 5$.
- `v3_paper_formulas.py`: the verifier's own transcription of the raw coefficients in (A.8), (A.10) and (A.12); (A.12) line 1 equals the corrected a_3, line 2 equals the published a_3, and every single sign change of the 12-term numerator destroys the first identity.
- `v4_appendixB.py`: claims B1–B3 above re-derived, plus B4 (prices 1).

See `verification.md` for the verdict and the list of corrections made to this manuscript.

4.4 Literature

No erratum or corrigendum is registered for [1] (Crossref record, checked 18 October 2026), and none of its 36 citing works indexed by OpenAlex corrects or proves the formula for $M_j \geq 3$. Later uses include the authors' crude-preheat study [2] (at most two exchangers per branch, unaffected), a simulation study of a real crude preheat network with 26 exchangers in four parallel branches that controls differences of Jäschke temperatures [3] (that network also has hot-stream interactions between branches, outside the paper's assumptions), and a thesis that re-implements the $M_1 = 4$ case with the published recursion [4]. Details are in `references.md`.

4.5 Domain of validity

- Theorem A is an identity of rational functions. It holds wherever the model is defined ($1 + wk_i \neq 0$, automatically true for positive parameters). There is no sign restriction: it also holds for exchangers that cool the branch, and in the unphysical AMTD regime $c_i > 1$ (small w).

- The temperature-only expression of c_i needs $\theta_{h,i} \neq \theta_{i-1}$. At $\theta_{h,i} = \theta_{i-1}$ the exchanger transfers no heat, yet $\partial Q_i/\partial w = a_i/2 = \frac{c_i}{2}(2\theta_{i-1} - A_{i-1})$ is generally nonzero, with $c_i = 2/(1 + wk_i)$ no longer measurable. So the paper's regularisation in §6.6.1 (setting $a_{i,j} = 0$ near this singularity) is itself only approximate.

- The model assumes that the hot-side data $(\theta_{h,i}, w_{h,i}, UA_i)$ do not depend on the split, i.e. no stream crossovers between branches. This is the same restriction as in the paper (§6.5).

- With LMTD and finite hot heat capacities, φ_i/ε_i depends on R_i as well as on ε_i. We have not proved that no exact formula in cold temperatures and hot inlet temperatures alone exists, but Proposition C gives an exact one that also uses the hot outlet temperatures. In the paper's §5 simulations the AMTD-based controlled variable is a (good) heuristic for LMTD plants. For $M_j \geq 3$ the corrected formula should be used.

4.6 The factor $-\frac{1}{2}$

- Literally, (16) and (19) are off by the factor $-\frac{1}{2}$ even for $M_j = 1$. For a single exchanger the correct relation is $\partial J_j/\partial w_j = -\theta_{1,j}^2/(2\theta_{h1,j})$.

- The sign comes from $J = -\sum pQ$, written as "$\partial J_j/\partial w_j = p_j\,\partial Q_j/\partial w_j$" in (A.6). The 2 comes from the mis-solved (A.5).

- The factor is the same for every branch, so the optimality condition "equal Jäschke temperatures" is unaffected. We regard this as a normalisation slip, not as part of the conjecture. Taken literally, however, (16) and (19) are false for every M_j, including the cases $M_j = 1, 2$ that the paper proves.

4.7 Points a referee should check first

(1) Lemma 1: the solution of the AMTD exchanger equations and the factor 2.

(2) Step 1, $\sum_{l \leq i} Q_l = w\theta_i$, which is the source of the cumulative sum.

(3) The comparison with the paper's (A.12): the transcription of the 12-term numerator from the rendered page 47 (claims S4 and P3–P6).

(4) That the paper's recursion really subtracts only a_{i-1}: (18), (29), (A.13), (A.15) and (B.1) all write it that way.

References

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